SSDT: Multiscale Spatial–Spectral Dilated Transformer for Hyperspectral and Multispectral Image Fusion

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***Abstract*—Hyperspectral images (HSIs) and multispectral image (MSI) possess complementary advantages, with HSIs providing rich spectral information and MSIs offering fine spatial details. Therefore, fusing HSIs and MSIs to obtain high-resolution HSIs (HR-HSIs) with both high spatial detail and rich spectral information has drawn increased attention. However, convolutional neural network (CNN)-based methods are limited by their local receptive fields, while traditional Transformer architectures suffer from extremely high computational complexity when applied to HSIs with numerous spectral bands. To address these issues, we propose a novel multiscale spatial–spectral dilated Transformer (SSDT) network based on the Transformer framework. The proposed method adopts a dual-branch architecture, consisting of the spatial dilated multiscale Transformer (Spa-DMST) and the spectral dilated positional embedded Transformer (Spe-DPET) modules. Specifically, the Spa-DMST introduces a multilevel dilated window mechanism to enable cross-scale spatial feature extraction, while the Spe-PEDT employs position encoding-enhanced dilated attention to reduce computational complexity. To further optimize the feature fusion process, we design a gating fusion reconstruction (GFR) module that integrates depthwise separable convolutions with a gating mechanism, effectively suppressing redundant information and enhancing detail reconstruction capabilities. Extensive experiments and visualized results on three simulated datasets and one real dataset demonstrate that the proposed method outperforms other state-of-the-art fusion methods. The code of this article will be available at https://github.com/FxyPd/SSDT**

***Index Terms*—Dilation, hyperspectral image (HSI), image fusion, multispectral image (MSI), Transformer.**

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I. Introduction

HYPERSPECTRAL image (HSI) is characterized by a large number of bands, narrow bandwidths, and extensive spectral coverage. The spectral range covers visible, infrared, and near-infrared, providing hundreds or even thousands of contiguous narrow bands [1], allowing for precise representation of different objects. The fine spectral detail significantly enhances accuracy in land-cover classification [2], target recognition [3], environmental monitoring [4], and resource surveys [5].

However, the inherent physical properties of the hyperspectral imaging sensors lead to a tradeoff between spatial and spectral resolution. Specifically, because the total energy of incoming photons is fixed during imaging, increasing the number of spectral bands inevitably weakens the spatial resolution of the image. Under the same total photon energy, the spatial resolution of the HSI is consistently lower than that of a multispectral image (MSI), limiting the usability of HSIs [6]. Therefore, when the low-resolution HSI (LR-HSI) and the high-resolution multispectral image (HR-MSI) can be simultaneously acquired, multispectral and HSI fusion (MHIF) techniques can be employed to leverage the strengths of both images [7], resulting in high-spatial-resolution HSIs that meet the needs of diverse applications.

Traditional methods for MHIF often enhance fusion performance by designing regularization conditions based on prior information about HSIs [8], [9], such as matrix factorization [10], [11] and tensor decomposition approaches [12], [13]. However, the effectiveness of these methods largely depends on the similarity between the specified prior knowledge and the fundamental characteristics of the data, which requires extensive expertise and can be time-consuming. Additionally, manually created priors may lead to incomplete feature extraction, ultimately affecting the fusion results. With the rapid development of deep learning, researchers have introduced convolutional neural network (CNN)-based methods into MHIF. For instance, the SSRNET by Zhang et al. [14] and the TFNet, along with its enhanced version ResTFNet, proposed by Liu et al. [15], have shown better results than traditional methods due to the strong feature inference and learning capabilities of CNNs. However, CNNs have the drawback of limited receptive fields, which hinders their ability

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to effectively capture global features of the image, thereby constraining fusion capabilities. In contrast, Transformer-based methods, which can capture long-range dependencies [16], have been successfully applied in the field of hyperspectral and multispectral image fusion, yielding promising results, as seen in works like Fusformer [17] and DCT [18].

Although deep learning methods based on CNNs and Transformers have achieved significant success in HSI and MSI fusion tasks, there still exist some issues: 1) although Transformer-based methods alleviate the limited receptive field constraint inherent in CNN architectures through global modeling capability, their computational complexity grows quadratically with the spectral dimension when processing HSIs containing hundreds of spectral bands and 2) most existing fusion approaches rely on simplistic cross-modal interaction strategies, where HSIs and MSIs are directly concatenated via channel stacking [17], [19], [20], [21] as model input. This will lead to feature redundancy during feature extraction and fusion, while ignoring the cross-modal information interaction between HSIs and MSIs, resulting in suboptimal fusion performance. Remote sensing images typically cover large areas with abundant structural information and texture features, which require the integration of global and local information at multiple levels to recover. Therefore, efficiently modeling the long-distance dependency of the image and multiscale information is of great importance for the MHIF task.

To address the above issues, this article proposes an HSI and MSI fusion based on the combination of a dilation mechanism and a Transformer. The proposed spatial–spectral dilated Transformer (SSDT) network consists of two stages: the first stage receives HSIs and MSIs as input to extract the spatial and spectral features required for the fusion stage, and the second stage aggregates and reconstructs the extracted features to generate the final output image. The first stage consists of the collaborated spatial dilated multiscale Transformer (Spa-DMST) and the spectral dilated positional embedded Transformer (Spe-DPET). By combining the dilation mechanism with multiscale information, this stage extracts global and local features at different scales to obtain richer multilevel information. In the second stage, we designed the gating fusion reconstruction (GFR) module, aiming to efficiently recover the multilevel spatial and spectral information of remote sensing images using multiscale features and a gating mechanism. Unlike the progressive framework proposed by [22], which leverages the VGGNet to extract multiscale features, our method employs a specially designed module to progressively extract hierarchical features during the feature extraction stage, thus providing a stronger capability in capturing global contextual information and a more fine-grained representation across different feature scales. The contributions of this article are as follows.

1) We propose a multiscale SSDT network for hyperspectral and multispectral image fusion. The SSDT consists of a multiscale feature extraction stage and a hierarchical fusion stage. This framework can capture global–local refined features and enable cross-modal interaction with redundancy suppression. Our experimental results on multiple datasets demonstrate that the proposed method achieves state-of-the-art (SOTA) performance.
2) Dilation mechanism is combined with the Transformer to effectively extract global–local refined spatial and spectral features at different scales. Specifically, we designed an Spa-DMST and an Spe-DPET to effectively extract multiscale spatial and spectral features, respectively, allowing the capture of larger receptive field information without losing the information in the local area.
3) We designed a deep fusion module using a gating mechanism for the final fusion stage, that is, the GFR module. In this process, we progressively integrate features from multiple scales to achieve more refined and detailed fusion results.

We organize this article as follows. Section II provides a brief overview of multimodal RS data fusion and attention mechanisms in DL. Section III presents the proposed model and algorithm in detail. Section IV describes the experimental setup and analysis of the results. Finally, Section V presents the conclusion of this article.

II. Related Work

In this section, we first introduce traditional methods for MHIF and then provide an overview of the applications of CNN and Transformer methods in image fusion in deep learning.

Currently, traditional methods for MHIF can be broadly categorized into matrix factorization and tensor factorization approaches. Matrix factorization assumes that the target HSI can be represented as a 2-D matrix, which is then decomposed into the product of two matrices for fusion. A common approach involves introducing point spread functions to establish the relationship between LR-HSIs and HR-MSIs; for example, Dian et al. [23] developed multiple prior constraints to optimize the cost function, achieving satisfactory fusion results. On the other hand, tensor factorization retains the original 3-D format of the data, maintaining spatial and spectral arrangements, which helps uncover internal relationships within the image. Notable methods include Tucker decomposition and canonical polyadic (CP) decomposition. Li et al. [12] proposed using sparse tensor representations to solve for the core tensor and dictionary for image fusion, while Xu et al. [24] clustered similar image patches and applied nonlocal CP decomposition for MHIF. Additionally, other traditional methods, such as wavelet transforms [25], have also been explored by researchers.

Considering that most traditional methods rely on manually set prior information to enhance fusion effects, which requires extensive expertise and does not fully leverage the complete features of the data, deep learning algorithms have emerged as a powerful alternative. These algorithms by deep learning can autonomously learn data features through multilayered structures and achieve efficient fusion without depending on prior knowledge. Researchers have introduced CNN-based methods for MHIF. For instance, Masi et al. [26] proposed a PNN using CNNs for image pansharpening, while Palsson et al. [27] developed a 3-D CNN method focusing on

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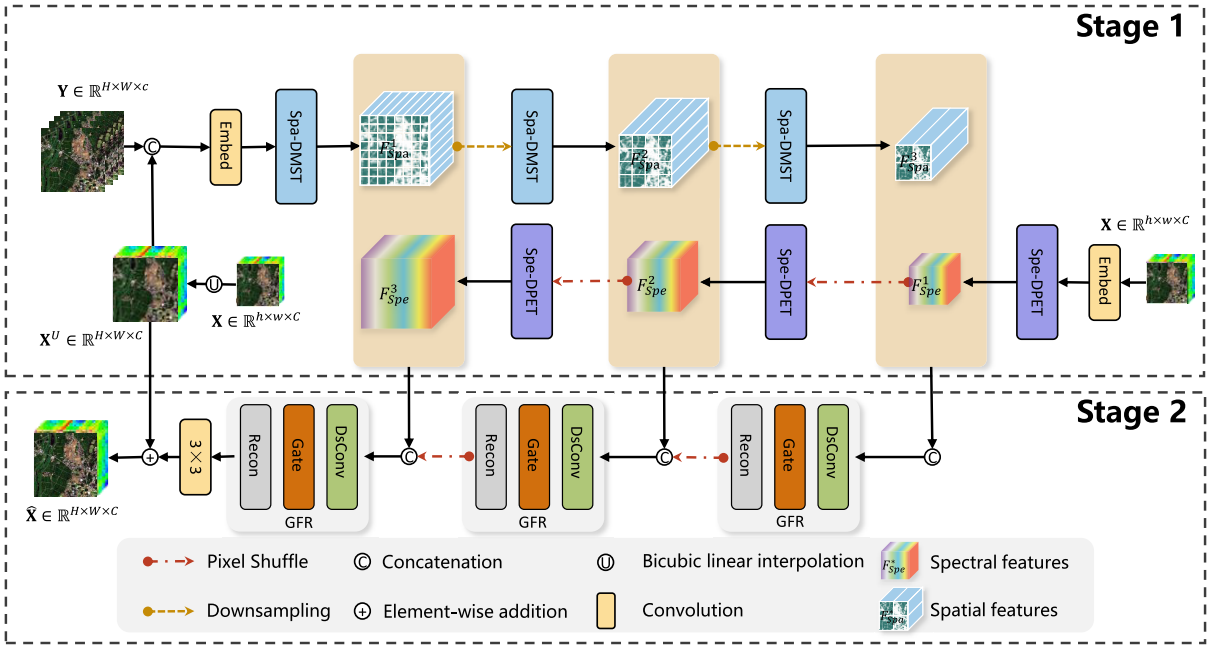


Fig. 1. Overall structure of the proposed SSDT network, which contains two stages. Stage 1 includes two branches: one for spatial feature extraction and the other for spectral feature extraction. Stage 2 is the fusion and reconstruction stage, which receives the outputs from Stage 1 as inputs. In this diagram, $\mathbf{X}$ represents the LR-HSI, $\mathbf{Y}$ represents the HR-multispectral image (HR-MSI), $\mathbf{X}^U$ denotes the LR-HSI after bicubic interpolation, and $\hat{\mathbf{X}}$ indicates the final fused result.

the spectral dimensions of HSIs, which achieved dual advantages by substantially lowering computational overhead while conferring enhanced noise tolerance. Additionally, Zhang et al. [14] introduced the SSR-NET framework, optimizing it with spatial and spectral edge loss. Huang et al. [28] combined deep spatial–spectral regularization with a physical imaging model in their DHIF for HSI fusion, but this method, based solely on convolutional operation, fails to adequately represent the complex degradation process, leading to certain distortions.

Despite the strengths of CNNs, their limited receptive field hinders the ability to capture global contextual information effectively. In many other classical tasks, such as object detection [29], [30], [31] and semantic segmentation [32], [33], researchers often employ downsampling strategies or use large strides in convolutions to increase the receptive field without significantly increasing computational costs [34], [35]. However, these methods result in reduced resolution of the feature maps during image feature extraction, which can negatively impact model performance. Early work by Chen et al. [36] introduced dynamic dilated convolutions (DDCs), which utilize dynamic weights to generate different convolution kernel parameters based on varying feature maps. Subsequently, Yan et al. [37] proposed a method using dilated convolutions, allowing for an increased receptive field without sacrificing image resolution by setting different dilation rates. For the MHIF tasks, Deng et al. [38] designed a dilation-based fusion network, BDT, and achieved promising results. However, BDT adopts only a single dilation rate in the spatial feature extraction stage, which may limit its ability to effectively capture global contextual information across the image.

The adaptation of Transformers into computer vision has yielded better results in image classification [39], object detection [40], and other CV tasks. It has also been confirmed that Transformers possess strong capabilities in modeling long-range dependencies. Consequently, researchers have applied Transformers to MHIF tasks. Hu et al. [17] first utilized Transformers in this context, proposing Fusformer to reconstruct high-quality HR-HSIs from LR-HSIs and HR-MSIs. Due to the extremely high number of spectral bands in HSIs, this approach results in excessive memory consumption and computational complexity. Wang et al. [21] introduced a multilevel cross-modal interaction module to extract multiscale features. Jia et al. [41] employed a self-supervised pretraining strategy in a dual-branch network to leverage Transformers for effective spectral and spatial feature extraction. Recently, Ma et al. [18] achieved outstanding results with their DC Transformer, which uses cross-attention for bidirectional information injection, but this approach also introduced significant computational complexity. We combined the concept of dilation with Transformer methods to propose the SSDT for hyperspectral and multispectral image fusion, which resulted in a notable performance improvement. For more details, refer to Section III.

III. Proposed Method

A. Overall Structure of the SSDT Network

The multiscale SSDT network for hyperspectral and multispectral image fusion proposed in this article is divided into two stages: the first stage extracts multiscale spatial and spectral features, and the second stage fuses the extracted features to reconstruct a fused image. The overall structure is shown in Fig. 1. In the first stage, the inputs are the LR-HSI and the HR-multispectral image (HR-MSI), denoted as

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$\mathbf{X} \in \mathbb{R}^{h\times w\times C}$ and $\mathbf{Y} \in \mathbb{R}^{H\times W\times c}$, respectively, where h and H represent the height, w and W represent the width, C and c represent the number of channels of LR-HSIs and HR-MSIs, respectively.

1) Stage 1: Feature Extraction: In the spatial feature branch of the first stage, we first apply bicubic interpolation to spatially upsample the LR-HSI, denoted as $\mathbf{X}^U \in \mathbb{R}^{H\times W\times C}$. Then, $\mathbf{X}^U$ and $\mathbf{Y}$ are concatenated along the channel dimension, followed by a convolution operation that adjusts the number of channels to the embedding dimension, which is 72 in our network. The output is then fed into the spatial feature extraction module Spa-DMST to obtain the spatial features $\mathbf{F}_{\text{spa}}$. To capture both the background information and fine texture details of the image, except for the first feature extraction block, a patch merging downsampling module is applied before each Spa-DMST block, reducing the resolution by half and doubling the number of channels to capture finer texture details. The following equation describes the process of the spatial feature extraction module:

$$\mathbf{F}^i_{\text{spa}} = \begin{cases} \text{Spa-DMST}\left(W_g\left(\text{Cat}\left(\mathbf{X}^U, \mathbf{Y}\right)\right)\right), & i = 1 \\ \text{Spa-DMST}\left(\text{Down}\left(\mathbf{F}^{i-1}_{\text{spa}}\right)\right), & i = 2, 3 \end{cases} \tag{1}$$

where $\mathbf{F}^i_{\text{spa}}$ $(i = 1, 2, 3)$ represents the spatial features at different scales. Spa-DMST$(\cdot)$ denotes the Spa-DMST module, Cat$(\cdot)$ represents the concatenation operation, $\text{W}_g(\cdot)$ is a 3×3 convolution used to adjust the number of channels, and Down$(\cdot)$ represents the patch merging downsampling operation.

The overall flow of the spectral feature branch is similar to the spatial feature branch. First, the LR-HSI is passed through a convolution operation to adjust the number of channels. The output is then passed to the spectral feature extraction module Spe-DPET, which extracts the spectral features $\mathbf{F}_{\text{spe}}$. Unlike the spatial branch, before each feature extraction block (except the first one), an upsampling operation is performed to better align the extracted features of the spectral feature branch. This process is described by the following equation:

$$\mathbf{F}^i_{\text{spe}} = \begin{cases} \text{Spe-DPET}\left(\text{W}_g\left(\mathbf{X}\right)\right), & i = 1 \\ \text{Spe-DPET}\left(\text{Up}\left(\mathbf{F}^{i-1}_{\text{spe}}\right)\right), & i = 2, 3. \end{cases} \tag{2}$$

Here, $\mathbf{F}^i_{\text{spe}}$ (where $i = 1, 2, 3$) represents the spectral features at different scales. Spe-DPET$(\cdot)$ denotes the spectral dilated positional embedded Transformer module and Up$(\cdot)$ represents the pixel shuffle upsampling operation.

2) Stage 2: Feature Fusion and Reconstruction: After feature extraction, the second stage fuses the spatial and spectral features to reconstruct the high-resolution HSI (HR-HSI) by the fusion module GFR. The feature fusion stage adopts a hierarchical fusion strategy, where each layer fuses spatial and spectral features at the corresponding scale. Additionally, except for the first layer, each layer receives the fusion result of the previous layer to fully utilize information at all scales. Since the fusion process starts with small-scale features and progresses to larger-scale features, an upsampling operation is applied before each layer (except the first) to match the size of the previous fusion result with features in the current layer. The second stage can be described as

$$\mathbf{Fus}_i = \begin{cases} \text{GFR}\left(\text{Cat}\left(\mathbf{F}^{4-i}_{\text{spa}}, \mathbf{F}^i_{\text{spe}}\right)\right), & i = 1 \\ \text{GFR}\left(\text{Cat}\left(\mathbf{F}^{4-i}_{\text{spa}}, \mathbf{F}^i_{\text{spe}}, \text{Up}\left(\mathbf{Fus}_{i-1}\right)\right)\right), & i = 2, 3 \end{cases} \tag{3}$$

where $\mathbf{Fus}_i$ $(i = 1, 2, 3)$ represents the fusion results at each scale. GFR$(\cdot)$ denotes the gating fusion reconstruction module. Finally, the fusion result $\mathbf{F}^3_{\text{fuse}}$ from the last layer is combined with the upsampled LR-HSI via an addition operation to generate the final output result.

B. Spatial Dilated Multiscale Transformer

After the introduction of Transformer models into MHIF tasks, many researchers employed a window partitioning strategy to reduce the high computational cost associated with global Transformer calculations [42], [43], [44]. However, limiting attention calculations to specific window regions restricts the ability to extract information from the window, which is disadvantageous for remote sensing images containing rich global correlation information [45]. To address this problem, we propose the Spa-DMST, as shown in Fig. 2. Compared with BDT, our approach does not rely on rearranging pixels to achieve spatial dilation. Instead, we adopt the concept of dilated expansion, employing multiple dilation rates to extract spatial features. Furthermore, we extend the dilation mechanism to the spectral feature extraction, which is overlooked by BDT. This design enables more comprehensive modeling of both global contextual relationships and local spatial details. This module divides the features into multiple groups along the channel dimension, and each group uses a different dilation rate to expand in the spatial dimension. This strategy allows the model to perceive local and global information while decreasing the computational complexity compared to window-based attention mechanisms, which will be validated in the ablation experiments in Section IV-E. The Spa-DMST consists of layer normalization (LN), dilated multiscale local self-attention (DMLA), and a multilayer perceptron (MLP). For input $\mathbf{X}$, the process is described by the following equation:

$$\mathbf{F}^M_{\text{spa}} = \text{DMLA}\left(\text{LN}\left(\mathbf{X}\right)\right) + \mathbf{X} \tag{4}$$

$$\mathbf{F}_{\text{spa}} = \text{MLP}\left(\text{LN}\left(\mathbf{F}^M_{\text{spa}}\right)\right) + \mathbf{F}^M_{\text{spa}} \tag{5}$$

where $\mathbf{X}$ represents the input features of the Spa-DMST module, $\mathbf{F}^M_{\text{spa}}$ represents the input of the MLP, and $\mathbf{F}_{\text{spa}}$ is the output of the Spa-DMST module.

To simultaneously capture both global background information and local texture information using different dilation rates, the DMLA first applies a 1×1 convolution to the normalized features and then reshapes the features into three groups. Each group is expanded using different dilation rates during the dilated self-attention (DSA) process. Specifically, for the DSA with a dilation rate j $(j = 1, 2, 3)$, the input feature is evenly divided into three parts along the channel dimension, namely $\mathbf{F}^j_Q \in \mathbb{R}^{H\times W\times d}, \mathbf{F}^j_K \in \mathbb{R}^{H\times W\times d}, \mathbf{F}^j_V \in \mathbb{R}^{H\times W\times d}$, where $d = C/3$. These parts are used to generate the query, key, and value matrices $\mathbf{Q}_j$, $\mathbf{K}_j$, and $\mathbf{V}_j$, respectively. The query $\mathbf{Q}_j$ is generated directly from $\mathbf{F}^j_Q$ via reshaping, while $\mathbf{K}_j$ and $\mathbf{V}_j$

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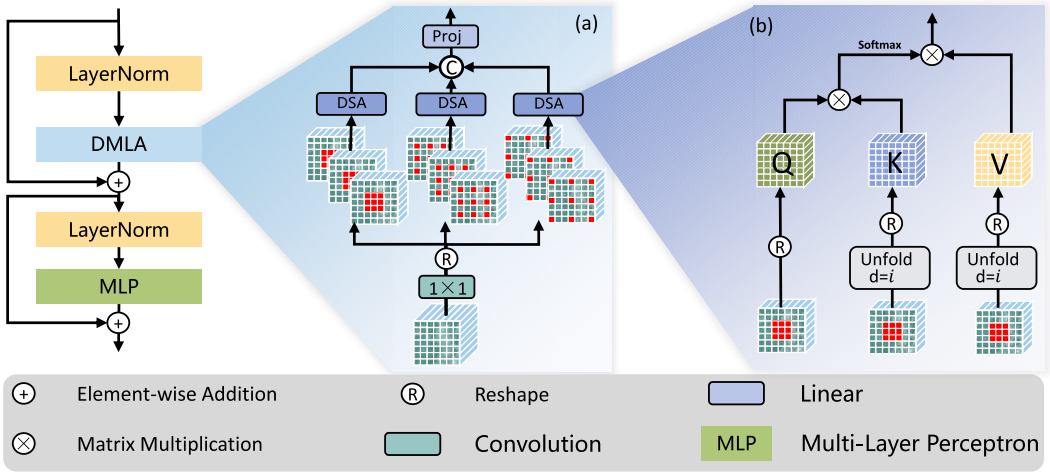


Fig. 2. Spatial feature extraction process, Spa-DMST. (a) Architecture of DMLA. (b) Process of DSA.

are generated by unfolding $\mathbf{F}_K^j$ and $\mathbf{F}_V^j$ with different dilation rates. This process is represented by the following equations:

$$\begin{cases} \mathbf{Q}_j = \text{Reshape}\left(\mathbf{F}_Q^j\right) \\ \mathbf{K}_j = \text{Reshape}\left(\text{Unfold}\left(\mathbf{F}_K^j\right)\right) \\ \mathbf{V}_j = \text{Reshape}\left(\text{Unfold}\left(\mathbf{F}_V^j\right)\right). \end{cases} \tag{6}$$

Next, the query, key, and value matrices undergo multihead attention calculation to obtain the feature information for dilation rate j, and the number of heads is H, which is 9 in our paper. The features obtained from different dilation rates are then concatenated and passed through a linear projection to integrate the multiscale receptive field information. The DMLA calculation process is as follows:

$$\mathbf{F}_j = \text{Cat}\left(\sigma\left(\frac{\mathbf{Q}_j^1\left(\mathbf{K}_j^1\right)^T}{\sqrt{d_M}}\right)V_j^1, \ldots, \sigma\left(\frac{\mathbf{Q}_j^H\left(\mathbf{K}_j^H\right)^T}{\sqrt{d_M}}\right)V_j^H\right) \tag{7}$$

$$\mathbf{F}_{\text{DMLA}} = \text{Linear}\left(\text{Cat}\left(\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3\right)\right). \tag{8}$$

In these equations, $\mathbf{F}_j$ represents the output of the DSA with dilation rate j, $\sigma(\cdot)$ denotes the Softmax activation function, and $\mathbf{F}_{\text{DMLA}}$ represents the output of the DMLA. $\mathbf{Q}_j^h$, $\mathbf{K}_j^h$, and $\mathbf{V}_j^h$ denote the query, key, and value metrics for the hth head ($h = 1, 2, \ldots, H$) and d_M represents the channel dimension of each head during multihead attention, $d_M = d/H$.

By setting different dilation rates in the DSA, the proposed Spa-DMST effectively integrates local texture features and nonlocal contextual information. It also reduces redundant information generated during attention computation, ensuring that as much information as possible is captured without increasing computational complexity.

C. Spectral Dilated Positional Embedded Transformer

Unlike traditional channel attention (CA) mechanisms that compress the entire feature map into a single vector through global pooling to calculate spectral features based on weights [46], [47], [48], the key to the spectral attention mechanism is to compute correlations between different spectral bands to capture deep spectral information [49]. However, the common approach of using the entire spatial range for feature map statistics presents two challenges: 1) high computational complexity when applied to remote sensing images with many channels and 2) high local spectral correlations might be overlooked in complex remote sensing scenes.

To address these issues, we propose an Spe-DPET module, as shown in Fig. 3. Unlike existing methods, we design a learnable channel positional encoder (CPE) before calculating spectral attention. This encoder learned the relative positional relationships of the spectra, thereby improving spectral feature extraction. Additionally, in the Spe-DPET module, to reduce the number of parameters, we employ two convolutions with shared weights in different branches and activation functions to design a dilator to generate query, key, and value pairs with different dilation rates. Both the Spe-PEDT and Spa-DMST are based on the Transformer framework, sharing similar structures. The key difference is that the Spe-PEDT utilizes dilated positional embedded attention (DPEA) after the first LN, while after the second LN, it uses a feed-forward neural network (FFN) instead of an MLP.

After the input is entered into each Spe-DPET module, the feature is first normalized using LN. Then, before computing CA, the designed CPE generates channel position encodings. The generated position encodings are finally added to the features after attention computation to capture the relationships between channels accurately. The position encoder we designed is learnable, making it adaptable to images with any number of channels and resolutions. It consists of two convolution operations and a GELU activation function.

For the feature tensor $\mathbf{F}_{\text{DPEA}}^{\text{in}}$, after LN, different dilators with various dilation rates were used to generate the query, key, and value pairs $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$

$$\begin{cases} \mathbf{Q} = \text{Dilator}_1\left(\mathbf{F}_{\text{DPEA}}^{\text{in}}\right) \\ \mathbf{K} = \text{Dilator}_2\left(\mathbf{F}_{\text{DPEA}}^{\text{in}}\right) \\ \mathbf{V} = \text{Dilator}_3\left(\mathbf{F}_{\text{DPEA}}^{\text{in}}\right) \end{cases} \tag{9}$$

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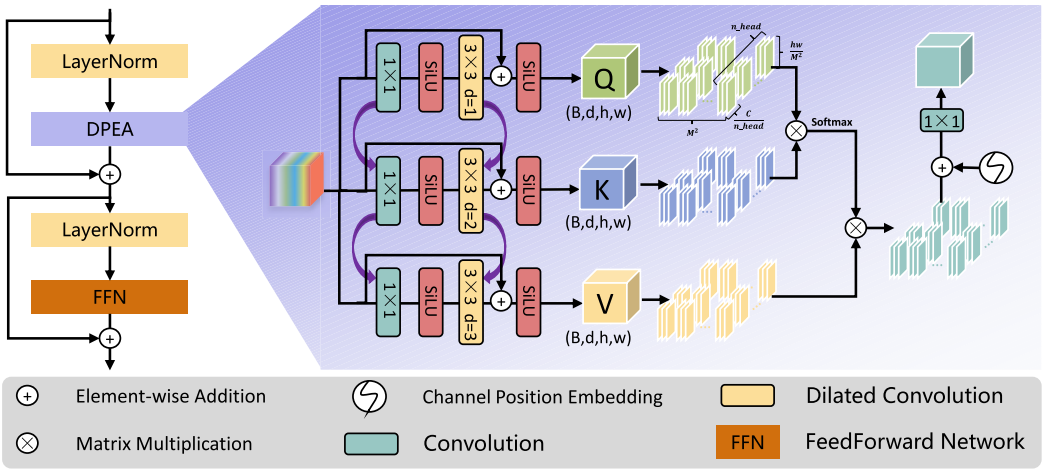


Fig. 3. Spectral feature extraction process, Spe-DPET. The subfigure at the right displays the core of the Spe-DPET, DPEA.

where $\mathbf{F}_{\text{DPEA}}^{\text{in}}$ represents the input features after LN. Dilator$_i$ means the dilation convolution, consisting of a 1×1 point-wise convolution, a 3×3 dilated convolution with a dilation rate i ($i = 1, 2, 3$), and two SiLU activation functions. The process for input $\mathbf{X}$ is described as

$$\text{Dilator}(\mathbf{X}) = \text{SiLU}\left(\mathbf{X} + W_g\left(\text{SiLU}\left(W_p(\mathbf{X})\right)\right)\right). \tag{10}$$

Here, W_p is the 1×1 convolutions in dilator. After that, $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ were reshaped and divided into M^2 groups $\mathbf{Q}_g$, $\mathbf{K}_g$, $\mathbf{V}_g \in \mathbb{R}^{M^2\times(HW/M^2)\times C}$, and grouped attention was calculated for the attention map $\mathbf{Attn}_i$ ($i = 1, 2, 3$), using the same method as described in 7. Finally, the attention results were added to the positional information, and a 1×1 convolution integrated the features to produce the spectral feature information

$$\mathbf{F}_{\text{spe}} = W_p\left(\mathbf{Attn}_i + \text{CPE}\left(\mathbf{F}_{\text{DPEA}}^{\text{in}}\right)\right) \tag{11}$$

where CPE represents the positional encoding operation. The proposed Spe-DPET module efficiently captures both global and local spectral correlations through the use of dilated convolutions and windowed attention, ensuring that key spectral features are not missed in complex remote sensing scenes while keeping computational complexity manageable. The learnable positional encoding further enhances the accuracy of interchannel relationships, improving spectral feature extraction.

D. GFR Module

The exploration of intrinsic spatial–spectral coherence in HSIs and MSIs is critical to effective fusion [22], [50]. To explicitly enhance the interaction between spatial and spectral features, the proposed GFR module, shown in Fig. 4, introduces a unified pathway that enables cross-domain feature exchange. The spatial and spectral features are first concatenated and passed through depthwise separable convolutions, which serve to preliminarily blend low-level structures and align them into a shared representation space. To further enable fine-grained spatial–spectral interaction, the fused features are then divided and processed by two parallel branches. Inspired by the gate mechanism in [49] and [51], each branch

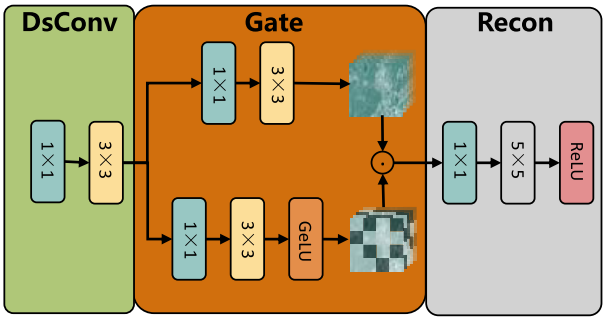


Fig. 4. Spatial and spectral feature fusion and the image reconstruction process, GFR.

applies depthwise separable convolutions independently, but with differing activation strategies, one incorporating a GELU nonlinearity. This design allows one branch to preserve the dominant information, while the other modulates it with contextual gating, effectively modeling interdependencies between spatial textures and spectral signatures. The outputs of both branches are combined via element-wise multiplication, which selectively emphasizes consistent and complementary features across the spatial and spectral domains. Finally, a large-kernel convolution is employed to consolidate the global context. Through this architecture, the GFR module facilitates explicit spatial–spectral feature interaction, leading to enhanced fusion performance. The process can be described as

$$\mathbf{Fus} = W_b\left(\text{Gate}\left(W_{\text{dp}}(\mathbf{F}_{\text{cat}})\right)\right) \tag{12}$$

where $\mathbf{F}_{\text{cat}}$ and $\mathbf{Fus}$ represent the concatenated future and the out of the GFR, respectively. $W_{\text{dp}}(\cdot)$ represents depthwise separate convolution, which consists of a pointwise convolution and a grouped convolution. $W_b(\cdot)$ refers to the large-kernel convolution and $\odot$ denotes element-wise multiplication. The gating mechanism is defined as

$$\text{Gate}(\cdot) = W_p\left(\text{Gelu}\left(W_{\text{dp}}^1(\cdot)\right) \odot W_{\text{dp}}^2(\cdot)\right). \tag{13}$$

Finally, the output of the last fusion module is further processed by a 3×3 convolution. This result is then added

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to the upsampled LR-HSI, which serves as a residual image, to produce the final HR-HSI. The final output is computed as follows:

$$\hat{\mathbf{X}} = \mathbf{X}^U + W_g\left(\mathbf{Fus}_{\text{out}}\right). \tag{14}$$

where $\mathbf{Fus}_{\text{out}}$ and $\hat{\mathbf{X}}$ denote the output of the last fusion layer and the proposed SSDT, respectively. Through these processes, the HR-HSI is reconstructed.

IV. Experiments

In this section, we will present experimental results on four datasets to analyze the effectiveness of the proposed method compared to eight state-of-the-art methods for hyperspectral and multispectral fusion.

A. Datasets

Three simulated datasets, namely Pavia University (PaviaU), Washington DC Mall (WDCM), and Chikusei, and a real dataset, the Yellow River Estuary (YRE) dataset, are tested in this article. The detailed information on these datasets is as follows.

1) *Pavia University:* The PaviaU dataset was captured by the Reflective Optics Spectrographic Imaging System (ROSIS-03) in 2003, over the city of Pavia, Italy. This spectrometer captures 115 continuous spectral bands in the wavelength range of 0.43–0.86 μm, with a spatial resolution of 1.3 m. However, due to noise interference, 12 bands were removed, leaving 103 usable spectral bands. The image size is 610×340 pixels. A 128×128 portion in the center of this cropped area was used as the test dataset, and a neighboring 128×128 portion to the right was used as the validation dataset.
2) *Washington DC Mall:* The HSI dataset of the WDCM was captured during the Hyperspectral Digital Imagery Collection Experiment (HYDICE) in 1995, using an airborne sensor over the National Mall in Washington, DC. The dataset originally contained 210 spectral bands, but after removing the bands affected by water vapor absorption, the number of valid bands was reduced to 191. The image has a pixel size of 1280×307 with a spatial resolution of 2.5 m. For data partitioning, two 128×128 sections were cropped from the lower left corner of the image to serve as the validation and test datasets, respectively.
3) *Chikusei:* The Chikusei dataset was captured using the Hyperspec-VNIR-CIRIS spectrometer during an aerial survey over Chikusei, Ibaraki, Japan. The dataset has a ground spatial resolution of 2.5 m with an image pixel size of 2517×2335. The bands of the dataset are 128. We selected a portion of the image with rich texture features, measuring 824×560 pixels, for the experiment. Two 256×256 regions were cropped from the lower-left corner for validation and testing, while the remaining area was divided into 64×64 patches for model training.
4) *Yellow River Estuary:* The YRE dataset is a full-resolution real remote sensing imaging dataset that comprises hyperspectral and multispectral images. The HSI of the dataset was acquired by the advanced hyperspectral imager (AHSI) onboard the Chinese Gaofen-5 satellite, while the multispectral image was captured by the MSI onboard the Sentinel-2A satellite. This data covers the eastern region of the Yellow River Delta in China. The hyperspectral imager provides spatial pixels measuring 1400×1400, with 280 spectral bands spanning wavelengths from 400 to 2500 nm. The multispectral image, with spatial pixels of 4200×4200, includes four spectral bands. The ground spatial resolution for the HSI is 30 m, while that of the multispectral image is 10 m. A region in the lower-left corner of the image was cropped for validation, and the remaining data was used as the training set for model development.

B. Experimental Settings

1) Data Simulation: For the three simulated datasets, that is, PaviaU, WDCM, and Chikusei, we used their original images as reference images, simulated the observed LR-HSIs and HR-MSIs by Wald's protocol [52], [53]. A Gaussian filter with a kernel size of 3×3 and a standard deviation of 2 was applied to the reference images to generate blurred images. These blurred images were then downsampled by a factor of 4 and 8 to create the simulated LR-HSI. Different spectral response functions were applied to the reference images to obtain HR multispectral images (HR-MSIs). For the real YRE dataset, during the training phase, we applied a Gaussian filter with a kernel size of 3×3 and a standard deviation of 2 to the HSI and downsampled it by a factor of 3, as done in [28], [54], to create the LR-HSI. The multispectral image was directly downsampled by a factor of 3 to create the HR-MSI, and the original HSI was used as a reference image for training. Once training is completed, the original hyperspectral and multispectral images are input into the trained model to generate the final fused image. For both the simulated and real datasets, we ensured that there was no overlap between the training, validation, and testing datasets.

2) Implementation Details: The SSDT model was trained on a 64-bit Linux system with an Intel[1] Xeon[1] Silver 4314 CPU running at 2.4 GHz and an Nvidia A40 GPU with 48GB of RAM. The entire experiment was conducted using Python version 3.8 and PyTorch version 1.12.1. During training, we set the number of training epochs to 2000 and used the AdamW optimizer with a learning rate of $1e^{-4}$. The size of the mini-batch is set to 32.

C. Compared Methods and Evaluation Metrics

Three state-of-the-art CNN-based methods, that is, SSR-Net [14], MoG-DCN [55], and DHIF [28], along with five Transformer-based methods, that is, PSRT [56], DSPNet [57], 3DT [20], MIMO [58], and CYformer [59], which explicitly modeled spatial–spectral interaction, are chosen as competitors. For all these methods, we partition the corresponding datasets into training and testing sets as described in

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TABLE I

PSNR, ERGAS, SAM, AND SSIM RESULTS OF DIFFERENT FUSION METHODS ON THREE SIMULATED DATASETS (THE BEST VALUES IN BOLD AND THE SECOND-BEST VALUES ARE UNDERLINED)

Ratio	Method	PaviaU				WDCM				Chikusei			
		PSNR	ERGAS	SAM	SSIM	PSNR	ERGAS	SAM	SSIM	PSNR	ERGAS	SAM	SSIM
4×	SSRNET	38.03	2.066	3.471	0.9707	40.84	2.005	1.758	0.9899	42.20	2.197	1.353	0.9889
	MoG-DCN	40.89	1.642	2.498	0.9841	41.91	1.645	1.550	0.9915	43.05	1.903	1.224	0.9914
	DHIF	41.49	1.516	2.331	0.9874	38.89	1.590	2.205	0.9905	43.69	1.744	1.137	0.9926
	PSRT	41.79	1.469	2.243	0.9861	37.02	2.363	2.729	0.9853	43.32	2.006	1.188	0.9919
	DSPNet	41.99	1.466	2.199	0.9865	41.59	1.779	1.614	0.9916	43.49	1.882	1.163	0.9922
	3DT	41.92	1.475	2.216	0.9868	42.06	1.760	1.531	0.9925	43.76	1.849	1.130	0.9924
	MIMO	42.47	1.387	2.077	0.9879	42.01	1.694	1.537	0.9922	43.87	1.845	1.106	0.9934
	CYformer	42.14	1.399	2.162	0.9874	42.93	1.509	1.384	0.9937	43.42	2.047	1.173	0.9924
	SSDT	**43.08**	**1.285**	**1.940**	**0.9895**	**44.14**	**1.252**	**1.204**	**0.9953**	**44.02**	**1.699**	**1.095**	**0.9933**
8×	SSRNET	37.89	2.085	3.523	0.9720	37.40	3.519	2.612	0.9881	39.21	3.157	1.906	0.9810
	MoG-DCN	39.10	1.945	3.056	0.9796	39.64	2.147	2.000	0.9880	39.49	3.115	1.802	0.9853
	DHIF	40.77	1.621	2.516	0.9837	39.96	2.002	1.948	0.9904	39.67	2.523	1.808	0.9883
	PSRT	39.76	1.788	2.840	0.9870	37.56	2.435	2.569	0.9890	39.30	3.330	1.883	0.9843
	DSPNet	39.91	1.809	2.793	0.9821	39.09	2.261	2.152	0.9884	40.31	2.936	1.664	0.9867
	3DT	41.02	1.600	2.456	0.9852	39.65	2.155	2.017	0.9887	40.51	2.849	1.641	0.9858
	MIMO	41.55	1.493	2.313	0.9861	40.71	1.882	1.786	0.9907	40.51	3.052	1.640	0.9875
	CYformer	41.41	1.475	2.350	0.9866	40.67	1.951	1.789	0.9909	40.38	3.077	1.665	0.9873
	SSDT	**41.98**	**1.385**	**2.122**	**0.9897**	**41.51**	**1.702**	**1.625**	**0.9941**	**40.85**	**2.402**	**1.576**	**0.9901**

Section IV-B. We then retrain these methods according to the descriptions in their respective papers under the same dataset settings, allowing us to compare their testing results with our proposed method in a fair environment.

For the three simulated datasets with reference images, we utilized four commonly used metrics: PSNR [60], ERGAS [61], SAM [62], and SSIM [63] to evaluate the performance of the proposed model in the fusion task. Since the YRE dataset consists of full-resolution remote sensing images, there is no reference image available to evaluate the quality of the fused results. Therefore, we adopted a commonly used no-reference metric [64], quality with no reference (QNR) [65], to evaluate the performance of the proposed model in real full-resolution scenarios.

D. Experimental Results

1) Results on Simulated Datasets: Table I displays the quantitative results of the SSDT and the compared methods on the PaviaU, WDCM, and Chikusei datasets under downsampling ratios of 4 and 8. The best results are highlighted in bold, while the second-best results are underlined. When the downsampling ratio is 4, MIMO achieves the second-best performance on most metrics for the PaviaU and Chikusei datasets. DHIF gets the second-best ERGAS value on Chikusei, while CYformer achieves the second-best performance across all metrics on WDCM. For the downsampling ratio of 8, MIMO and CYformer alternately achieve the second-best scores on all metrics for PaviaU and WDCM. On the Chikusei dataset, MIMO, 3DT, and DHIF each attain second-best results on different metrics. It is worth noting that the multiscale-based MIMO method and CYformer, which models spatial–spectral interaction, demonstrate strong performance across all datasets under both downsampling scales. However, our proposed SSDT, adopting a progressive scale-reduction strategy during feature extraction, consistently outperforms all the methods, achieving the best overall results with significant improvements across all datasets and downsampling ratios. Furthermore, the SSDT enhances both global context modeling and the preservation of local salient details, thereby achieving a better balance between global consistency and local fidelity, which ultimately leads to superior fusion results.

Figs. 5–7 illustrate the fusion results and the corresponding residual maps for each method. The residual maps are computed as the mean absolute error between the predicted image and the reference image across all spectral bands. In these maps, darker regions indicate higher similarity to the reference image, while brighter regions reflect greater deviations. Fig. 5 shows the pseudo-color fusion results and residual maps on the PaviaU dataset under different downsampling ratios. It is evident that the SSDT produces results with the highest similarity to the reference image, confirming its superiority over the compared methods on this dataset. Fig. 6 displays the qualitative results on the WDCM dataset. When the downsampling ratio is 8, most comparison methods suffer from severe color distortions and poor spatial detail reconstruction. In contrast, the SSDT demonstrates superior performance in both structural detail and spectral fidelity, further validating its effectiveness on this dataset. Fig. 7 shows the fusion results on the Chikusei dataset. Similar to the observations on the previous two datasets, the SSDT consistently exhibits excellent reconstruction quality. Overall, the proposed method achieves better spatial information preservation in the red boxed regions across all three simulated datasets compared with other competing methods, highlighting the strong reconstruction capability of the SSDT.

To further validate the reconstruction performance of the proposed method at the spectral level, we present the per-band PSNR values on the three simulated datasets under downsampling ratios of 4 and 8. These values are calculated

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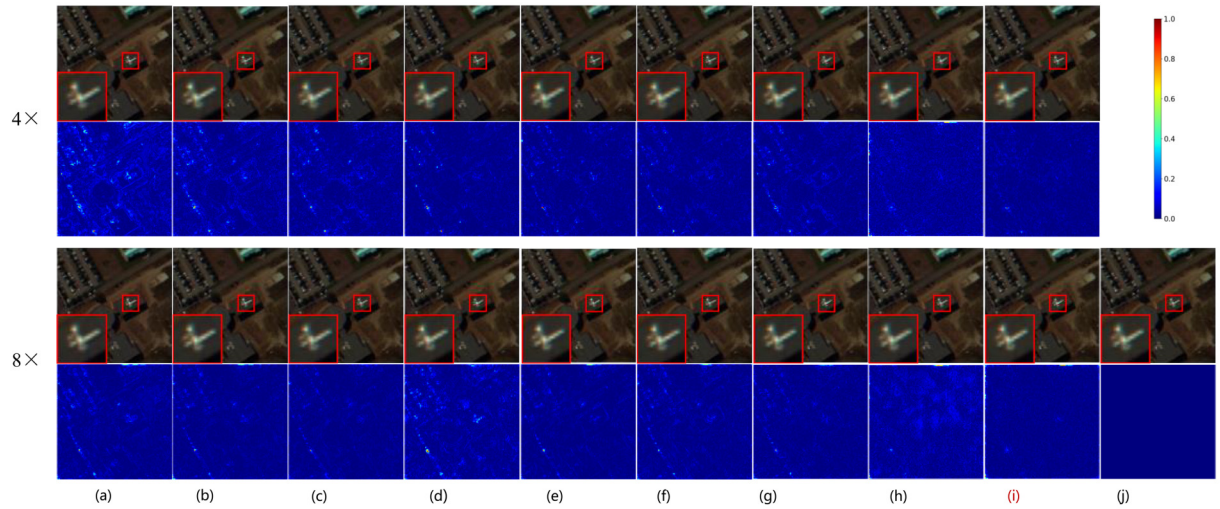


Fig. 5. Visualization of the fusion results of different methods on the PaviaU dataset. The first and third rows display the pseudo-color images of the fusion results for each method, corresponding to downsample ratios of 4× and 8×, and using RGB bands 67, 29, and 6, respectively. The second and fourth rows show the residuals between the reference image and the fusion results. (a) SSRNet. (b) MoGDCN. (c) DHIF. (d) PSRT. (e) DSPNet. (f) 3DT. (g) MIMO. (h) CYformer. (i) SSDT. (j) GT.

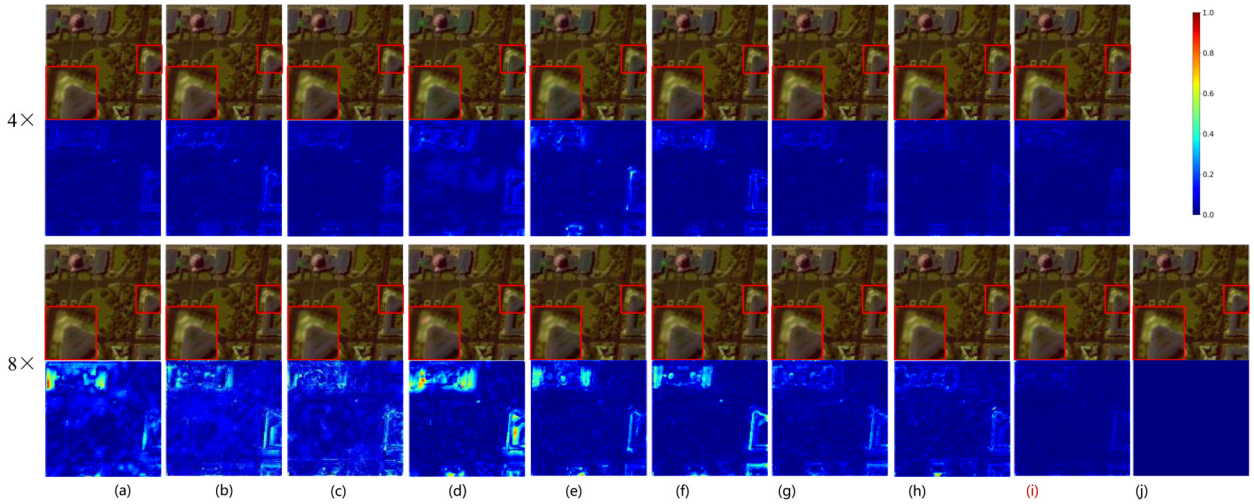


Fig. 6. Visualization of the fusion results of different methods on the WDCM dataset. The first and third rows display the pseudo-color images of the fusion results for each method, corresponding to downsample ratios of 4× and 8×, and using RGB bands 55, 120, and 10, respectively. The second and fourth rows show the residuals between the reference image and the fusion results. (a) SSRNet. (b) MoGDCN. (c) DHIF. (d) PSRT. (e) DSPNet. (f) 3DT. (g) MIMO. (h) CYformer. (i) SSDT. (j) GT.

by comparing each band of the reconstructed results with the corresponding band of the reference image. The results are illustrated in Fig. 8. It is evident that the SSDT achieves the highest PSNR values across most spectral bands, with only a few bands slightly outperformed by some other methods, which proves the superior overall reconstruction quality compared to all other approaches.

2) Results on the PaviaU Dataset With Different Noises: To account for the inevitable presence of noise in real remote sensing images, we conducted additional comparative experiments under different noise conditions on the PaviaU dataset using a downsampling ratio of 4. Specifically, the LR-HSI input was corrupted with Gaussian noise at 20, 30, and 40 dB, respectively, while Gaussian noise with a fixed SNR of 40 dB was added to the HR-MSI.

The quantitative results are summarized in Table II. It can be observed that the SSDT achieves the best performance across all noise levels. CYformer performs second-best under 20-dB noise, while MIMO ranks second under 30- and 40-dB conditions. To complement the quantitative evaluation, we also provide qualitative analysis through visualization of the pseudo-color fusion results and the corresponding residual maps under different noise conditions, as shown in Fig. 9. The results reveal that most competing methods suffer from significant performance degradation in the presence of noise. For instance, 3DT produces noticeable color distortions, and the reconstruction in the red-boxed region is not satisfactory. In contrast, the SSDT maintains superior performance even under severe noise conditions, showing the highest visual reconstruction effect. Overall, both quantitative and qualitative

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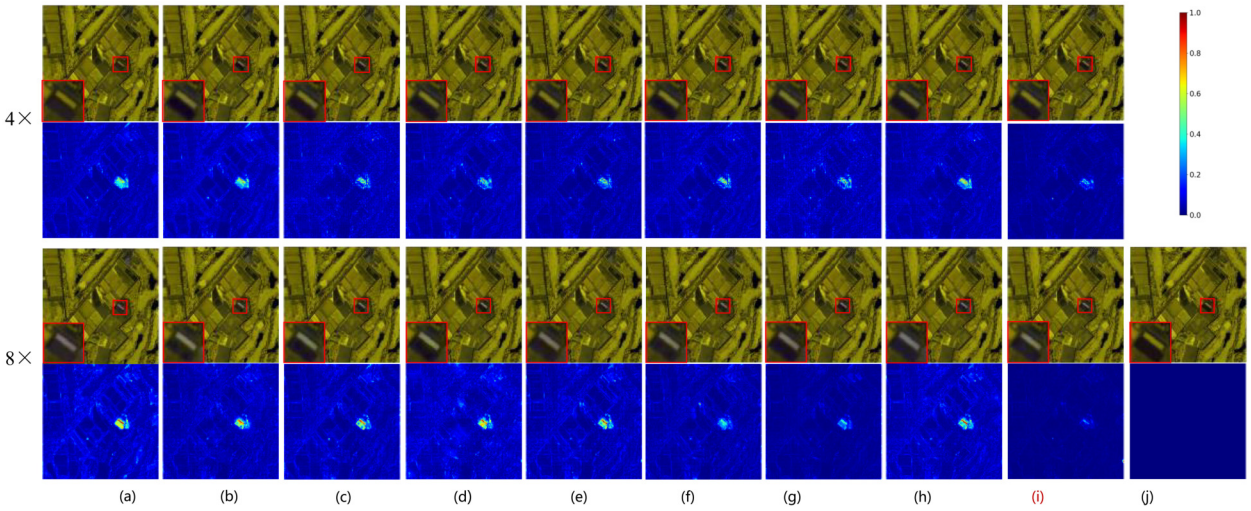


Fig. 7. Visualization of the fusion results of different methods on the Chikusei dataset. The first and third rows display the pseudo-color images of the fusion results for each method, corresponding to downsample ratios of 4× and 8×, and using RGB bands 80, 76, and 55, respectively. The second and fourth rows show the residuals between the reference image and the fusion results. (a) SSRNet. (b) MoGDCN. (c) DHIF. (d) PSRT. (e) DSPNet. (f) 3DT. (g) MIMO. (h) CYformer. (i) SSDT. (j) GT.

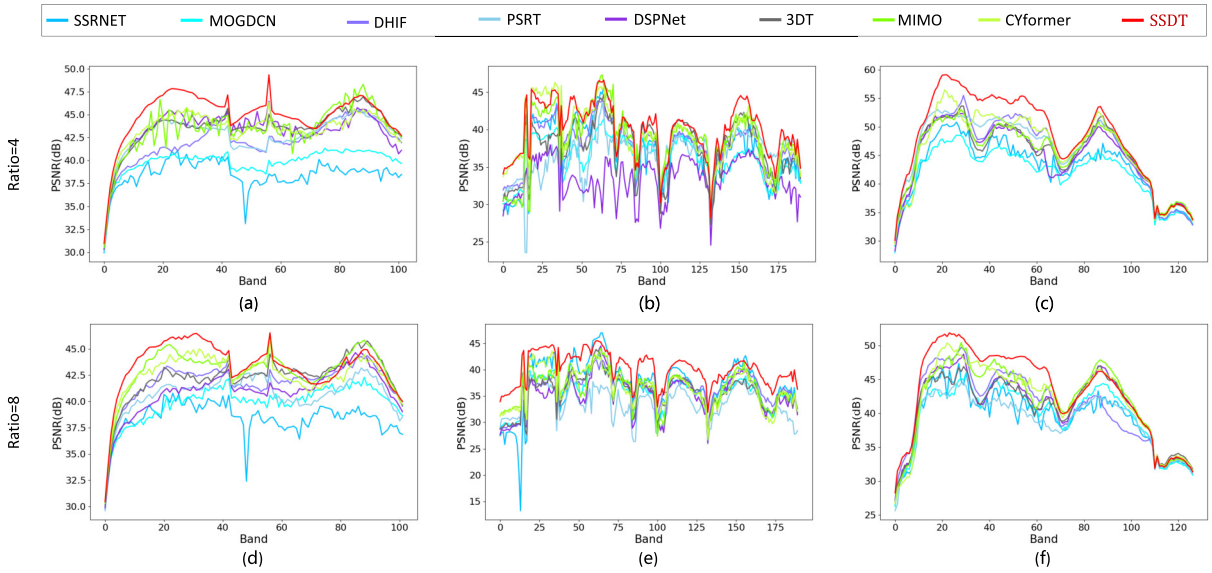


Fig. 8. PSNR values for different methods' fusion results across different bands. (a)–(c) PSNR values for the PaviaU, WDCM, and Chikusei datasets of the downsampling ratio 4, respectively. (d)–(f) PSNR values of the PaviaU, WDCM, and Chikusei datasets of the downsampling ratio 8.

TABLE II

PSNR, ERGAS, SAM, AND SSIM RESULTS OF DIFFERENT FUSION METHODS UNDER DIFFERENT GAUSSIAN NOISE LEVELS ON THE PAVIAU DATASET (THE BEST VALUES IN BOLD AND THE SECOND-BEST VALUES ARE UNDERLINED)

Method	20dB				30dB				40dB			
	PSNR	ERGAS	SAM	SSIM	PSNR	ERGAS	SAM	SSIM	PSNR	ERGAS	SAM	SSIM
SSRNET	27.85	6.021	11.17	0.7264	27.86	6.022	11.16	0.7260	27.86	6.022	11.16	0.7261
MoG-DCN	28.44	5.620	10.43	0.7658	30.04	4.646	8.658	0.8380	26.98	6.435	12.36	0.6996
DHIF	29.54	4.737	9.184	0.8384	25.52	7.486	14.69	0.5906	28.47	5.352	10.40	0.800
PSRT	34.65	2.950	5.085	0.9395	34.50	2.979	5.176	0.9384	34.27	3.049	5.311	0.9352
DSPNet	31.51	3.956	7.314	0.8913	31.53	3.901	7.291	0.8869	31.82	3.806	7.054	0.8961
3DT	26.98	6.864	12.38	0.6895	27.10	6.742	12.19	0.7010	27.04	6.787	12.27	0.7005
MIMO	35.86	2.600	4.418	0.9516	36.09	2.542	4.296	0.9547	36.24	2.496	4.231	0.9552
CYformer	35.96	2.628	4.351	**0.9610**	36.01	2.618	4.337	**0.9625**	35.95	2.635	4.371	**0.9617**
SSDT	**36.89**	**2.268**	**3.910**	0.9609	**37.11**	**2.260**	**3.824**	0.9599	**36.99**	**2.247**	**3.861**	0.9611

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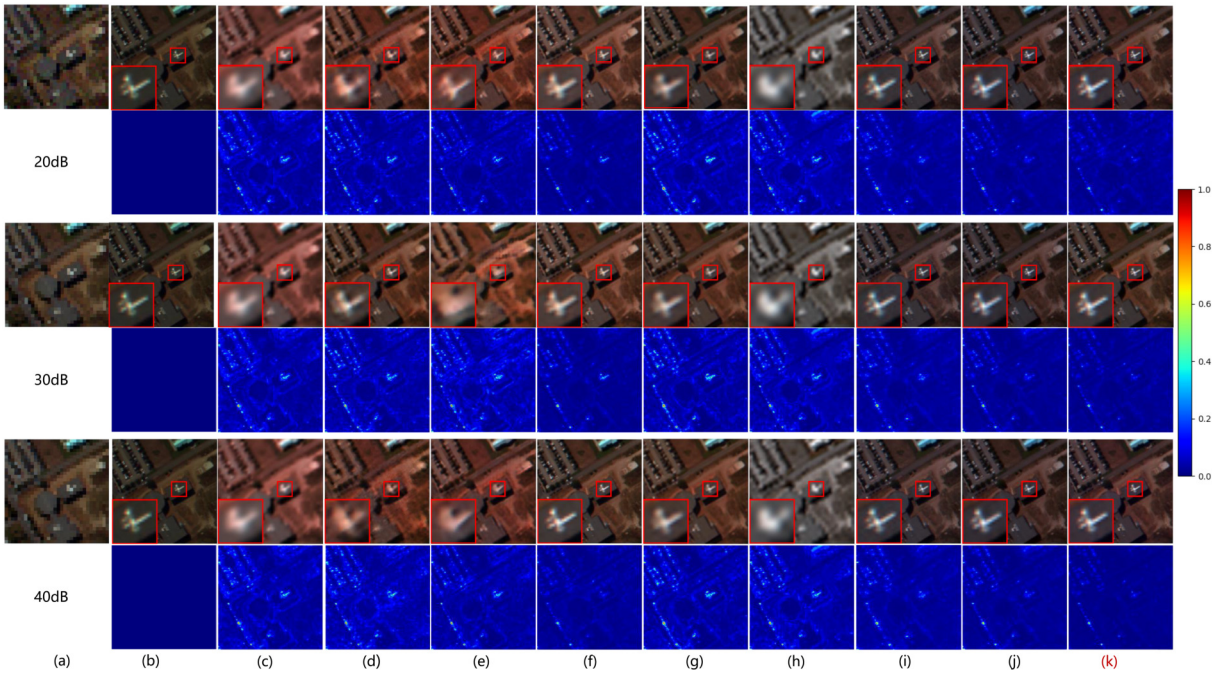


Fig. 9. Visualization of the fusion results under different Gaussian noise levels on the PaviaU dataset. The first, third, and fifth rows display the pseudo-color images of the fusion results from various methods with added Gaussian noise levels of 20, 30, and 40 dB, respectively. The pseudo-color images are generated using bands 67, 29, and 6. The second, fourth, and sixth rows present the corresponding residual images between the reference image and each fusion result. (a) LRHSI. (b) GT. (c) SSRNet. (d) MoGDCN. (e) DHIF. (f) PSRT. (g) DSPNet. (h) 3DT. (i) MIMO. (j) CYformer. (k) SSDT.

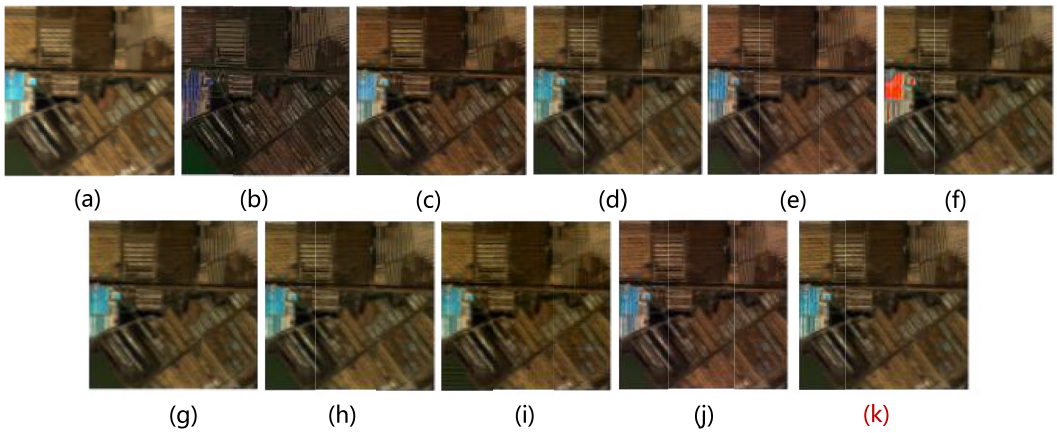


Fig. 10. Pseudo-color images of the LR-HSI, HR-MSI, and fusion results for each method of the YRE dataset, using RGB bands 80, 76, and 55. (a) LRHSI. (b) HRMSI. (c) SSRNet. (d) MoGDCN. (e) DHIF. (f) PSRT. (g) DSPNet. (h) 3DT. (i) MIMO. (j) CYformer. (k) SSDT.

analyses confirm that the proposed method exhibits robust and consistent fusion performance under varying levels of Gaussian noise, demonstrating its strong generalization and noise resistance capabilities.

3) Results on the Real Dataset: To better validate the effectiveness of the proposed method, we conducted comparative experiments on the YRE dataset and visualized the results. Fig. 10 displays the fusion results of various methods applied to the test set from the real dataset. The fusion results of the SSDT show colors that are closer to the LR-HSI images captured by the GF5 satellite, with overall textures being more pronounced and land cover edges clearer, demonstrating a distinct advantage over other methods. Since the YRE dataset lacks a reference ground-truth image, we employed the QNR metric to analyze the outcomes of our method against others. Fig. 11 illustrates the QNR metrics and the spatial and spectral distortion degrees for each method. The SSDT achieved the best results, indicating superior capabilities in recovering both spatial and spectral details.

4) Complexity Comparison of Different Methods: To further compare the computational complexities of various methods, we take the PaviaU dataset as an example to analyze the PSNR, the number of parameters, and the number of operations of different methods. The results are shown in Fig. 12, where the bubble size represents the parameter count, the horizontal axis denotes computational complexity, and the vertical axis reflects model performance (PSNR). As shown in the figure, the SSDT achieves the best

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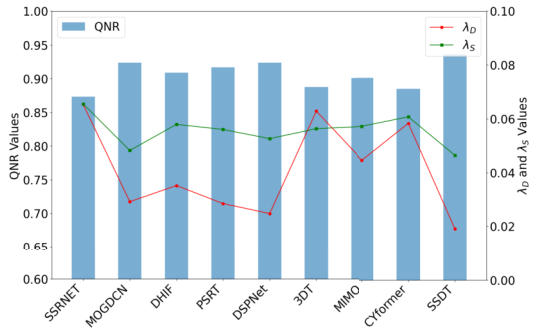


Fig. 11. QNR, λ_S, and λ_D values for the results of different methods on the YRE dataset. The bar chart represents the QNR values, with the y-axis scale on the left side of the figure, while the green and red curves show the values of λ_S and λ_D, with the y-axis scale on the right side of the figure.

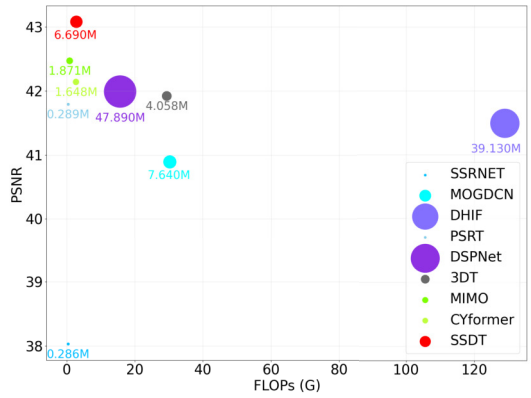


Fig. 12. Bubble chart of parameter count, computational complexity, and performance of different methods.

performance with a small FLOP and number of parameters, demonstrating optimal overall effectiveness. PSRT, due to its minimal parameter usage, achieves the lowest parameter count and computational complexity but slightly compromises performance. CNN-based methods, such as SSRNET, also feature small parameter counts and low computational complexity but exhibit significant performance shortcomings. In summary, the proposed SSDT strikes an optimal balance by achieving the best performance with moderate parameter usage and computational complexity, ensuring superior efficiency. To further validate the computational complexity differences among the methods, we compared their inference time during testing. Specifically, each method was executed ten times on the test sets of different datasets, and the average runtime was reported to minimize the influence of randomness. The final outcomes are presented in the corresponding Table III. As shown in the table, our method (SSDT) achieves competitive quantitative performance across all datasets. Notably, on the PaviaU and WDCM datasets, the SSDT ranks second-best value of 2.948 and 3.562, and on the Chikusei dataset, it also performs competitively with a score of 4.739. Although not the absolute best in terms of individual metric values, the SSDT maintains consistent performance across different datasets. These findings confirm that the proposed method achieves a well-balanced tradeoff between fusion accuracy and computational efficiency, maintaining reliable performance without incurring excessive inference cost.

TABLE III

COMPARISON OF AVERAGE INFERENCE TIME (IN SECONDS) OF DIFFERENT METHODS ON THREE DATASETS UNDER 4× DOWNSAMPLING RATIO (THE BEST VALUES IN BOLD AND THE SECOND-BEST VALUES ARE UNDERLINED)

Method	PaviaU	WDCM	Chikusei
SSRNET	**2.875**	3.790	<u>4.562</u>
MoG-DCN	3.072	4.070	4.888
DHIF	4.466	5.994	5.335
PSRT	4.365	4.819	4.627
DSPNet	2.979	3.609	4.621
3DT	3.151	3.961	6.051
MIMO	2.997	**3.407**	**4.476**
CYformer	3.532	4.515	4.783
SSDT	<u>2.948</u>	<u>3.562</u>	4.739

TABLE IV

ABLATION STUDIES OF THREE MODULES ON THREE DATASETS (THE BEST VALUES ARE HIGHLIGHTED IN BOLD)

Dataset		w/o Spa-DMST	w/o Spe-DPET	w/o GFR	SSDT
PaviaU	PSNR	40.59	42.96	41.96	**43.08**
	SAM	2.584	1.964	2.201	**1.940**
WDCM	PSNR	43.68	43.57	42.89	**44.14**
	SAM	1.267	1.287	1.395	**1.204**
Chikusei	PSNR	43.67	43.87	43.43	**44.02**
	SAM	1.140	1.115	1.160	**1.095**

E. Ablation Analysis

In this section, we first validate the effectiveness of the proposed modules on three simulated datasets from three perspectives: the effectiveness of the Spa-DMST, the effectiveness of the Spe-DPET, and the effectiveness of gating fusion. Table III displays the PSNR and SAM values corresponding to the configurations without the Spa-DMST, Spe-DPET, and gating fusion, respectively. In the analysis, we refer to the three network configurations, without the Spa-DMST module, without the Spe-DPET module, and without the gating fusion module, as "w/o Spa-DMST," "w/o Spe-DPET," and "w/o GFR," respectively. Subsequently, we calculated and analyzed the number of parameters and model complexity by replacing the dilation mechanism with traditional convolutions. Finally, we analyzed the effect of the number of feature scales and dilation rates used in the model.

1) Effectiveness of the Spa-DMST: To validate the overall impact of the proposed Spa-DMST, we removed the core DMST for spatial information extraction from the model and used the PSNR and SAM values to assess the module's effectiveness across the three datasets. The first column of Table IV shows the PSNR and SAM results for configurations without the Spa-DMST on the different datasets. The results indicate that removing the spatial expansion mechanism while keeping other modules unchanged led to a decline in these two

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TABLE V
ABLATION STUDY ON DIFFERENT RATES AND SCALES

Scale	D1	D2	D3	PaviaU			
				PSNR	ERGAS	SAM	SSIM
3	✗	✓	✓	42.59	1.402	2.076	0.9880
3	✓	✗	✓	42.62	1.396	2.040	0.9883
3	✓	✓	✗	42.51	1.415	2.108	0.9879
1	✓	✓	✓	42.55	1.371	2.060	0.9893
2	✓	✓	✓	42.86	1.312	1.991	0.9895
3	✓	✓	✓	**43.08**	**1.285**	**1.940**	**0.9895**

TABLE VI
ABLATION STUDIES OF COMPLEXITY ON THREE DATASETS (THE BEST VALUES ARE HIGHLIGHTED IN BOLD)

Dataset		3×3	5×5	7×7	SSDT
PaviaU	PSNR	42.91	42.89	42.50	**43.08**
	Params(M)	7.35	8.51	10.25	**6.69**
	Flops(M)	1.21×10^5	1.32×10^5	1.48×10^5	$\mathbf{2.87}\times10^4$
WDCM	PSNR	43.89	43.73	43.76	**44.14**
	Params(M)	9.37	10.54	12.28	**8.72**
	Flops(M)	2.48×10^5	2.61×10^5	2.76×10^5	$\mathbf{6.11}\times\mathbf{10^4}$
Chikusei	PSNR	43.74	43.62	43.39	**44.02**
	Params(M)	7.81	8.97	10.71	**7.15**
	Flops(M)	6.04×10^5	6.44×10^5	7.08×10^5	$\mathbf{1.44}\times\mathbf{10^5}$

metrics. Without the expansion mechanism, the network may fail to capture more contextual information from the images, which limits the fusion effectiveness of the model to some extent.

2) Effectiveness of the Spe-DPET: To assess the effectiveness of the proposed Spe-DPET, we replaced it with the CA module [46]. To enhance spectral information extraction, we added a residual connection to the CA module [66], [67] in our ablation experiments. The results across the three datasets are presented in the second column of Table IV. The findings indicate that after replacing the Spe-DPET, the fusion performance across all datasets weakened, demonstrating that our Spe-DPET outperforms the traditional CA module in extracting spectral information.

3) Effectiveness of GFR: To evaluate the effectiveness of the proposed GFR module, we conducted an ablation experiment by removing the entire GFR module while keeping the spatial and spectral feature extraction modules unchanged. In this variant, the final fusion was performed simply by concatenating the extracted features, followed by a single convolutional layer to align the number of output channels with that of the reference image. The corresponding results are presented in the third column of Table IV. As observed, the removal of the GFR module leads to a significant performance drop across all three datasets. These results demonstrate that the proposed GFR module effectively facilitates the interaction between spatial and spectral features, thereby contributing to superior fusion and reconstruction.

4) Impact of Feature Scales and Dilation Rates: We investigated the impact of feature scales and dilation rates used in the SSDT on fusion performance. Specifically, we evaluated the effect of the dilation rates employed in the spatial feature and spectral feature extraction modules, where "D1," "D2," and "D3" denote the exclusion of feature extractors with dilation rates of 1, 2, and 3, respectively, while keeping the other dilation rates unchanged. The corresponding quantitative results are summarized in Table V. As observed, removing any of the dilation rates leads to a decline in reconstruction performance, while the best results are achieved when all three dilation rates are used simultaneously, highlighting the benefit of multidilation feature extraction. Moreover, we assessed the influence of using different numbers of feature scales in the feature extraction stage. We compared the performance of the SSDT when using 1, 2, or 3 feature scales. The results indicate that using only a single scale results in a significant drop in fusion quality. Incorporating two scales improves performance to some extent, but the optimal results are consistently achieved when all three scales are used. These findings demonstrate that both the use of diverse dilation rates and the multiscale feature extraction mechanism contribute positively and complementarily to the overall fusion performance of the proposed model.

5) Complexity Analysis: To validate that the dilation mechanism can expand the model's information extraction scope and improve performance while decreasing the number of parameters and computational complexity, we replaced the dilation module with 3×3, 5×5, and 7×7 convolutions, corresponding to dilation rates of 1, 2, and 3, respectively, to generate the query, key, and value pairs used in attention computation. The resulting PSNR values, parameter counts, and computational complexities are shown in Table VI. The results indicate that the dilation mechanism enables the model to extract image features more effectively, achieving optimal feature representation with minimal parameter count and computational complexity. When replacing the dilation mechanism with standard convolutions, computational complexity increases significantly, with the 3×3 convolution yielding the second-best results. This observation was validated across three datasets. These findings demonstrate the effectiveness and efficiency of the dilation mechanism combined with the Transformer in multiscale spatial–spectral domains.

V. CONCLUSION

This article introduces a multiscale SSDT network designed for the fusion of HR-multispectral images and LR-HSIs. The SSDT network is comprised of two stages: a multiscale feature extraction stage and a hierarchical fusion reconstruction stage. It includes specialized modules for spatial and spectral feature extraction, namely the Spa-DMST module and the Spe-DPET module. These modules are adept at capturing local and global features without a quadratic increase in computational complexity. In addition, we incorporate depthwise separable convolutions and a gated fusion mechanism to refine the fusion process. The model was evaluated on four datasets, demonstrating state-of-the-art performance in hyperspectral and multispectral image fusion.

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However, although the proposed method achieves highly competitive performance, its multibranch and multistage architecture inevitably introduces additional computational overhead. While we have made efforts to control the model complexity during the design process, the overall computational cost and parameter count remain higher than those of some existing methods. In future work, we plan to reduce the redundancy in the model by optimizing the parameter configuration and exploring a more lightweight architecture. Moreover, considering the difficulty of obtaining well-registered reference images in real-world scenarios, future research could focus on developing unsupervised training frameworks to enhance the practicality and applicability of the model. Combining these two directions, future research is expected to broaden the applicability of the proposed method to a wider range of real-world fusion scenarios.

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